

CSMFAB000000000113-Vol-24

Safety, Legal, and Maritime Framework: Operating in International Waters

Series: CSMFAB000000000113 | Volume: 24 of 29 | June 2026

1. Legal Framework: International Waters

1.1 UNCLOS Definition of International Waters

Per the United Nations Convention on the Law of the Sea (UNCLOS, 1982): - Territorial sea: 12 nautical miles (22.2 km) from baseline - Exclusive Economic Zone (EEZ): 200 nm (370 km) — resource rights only, freedom of navigation - High Seas: beyond EEZ — no state jurisdiction

AMSR-UBSCLS operating zone: > 12 nm offshore = no territorial jurisdiction over the launch. Operations in EEZ: legal under UNCLOS Article 58 (freedom of navigation + scientific research).

1.2 U.S. Regulatory Analysis

FAA Space Launch Regulation: FAA AST (Office of Commercial Space Transportation) regulates under 14 CFR Part 450: - Applies to: launches from U.S. territory, by U.S. citizens, or using U.S.-registered vehicles - International waters with foreign-flagged vessels and non-U.S. citizens: FAA Part 450 likely inapplicable - Recommendation: Use Marshall Islands- or Panama-flagged vessels for maximum legal clarity

FCC Frequency Authorization: LoRa 915 MHz: Part 15 device — no individual license in U.S. In international waters: ITU Radio Regulations govern; LoRa at 915 MHz requires coordination but no individual license for low-power devices.

1.3 NOTAM/NOTMAR Filing

Aviation safety: file NOTAM (Notice to Airmen) specifying: - Launch location (lat/lon) - Launch window (date/time UTC) - Trajectory corridor (0-200 km altitude, ± 50 km lateral)

Maritime safety: file NOTMAR (Notice to Mariners) via USCG or regional authority.

Both filings are voluntary for international operations but strongly recommended for safety and liability protection.

2. Safety Zones

2.1 Minimum Exclusion Zones

Scale	H_apogee	Downrange (km)	Exclusion Radius	Aviation Alert Alt
Nano	5 km	10	200 m maritime, N/A aviation	Below FL180 — ATC notification
Micro	25 km	50	500 m maritime, 1 km aviation	Below FL650 (above airliners)
Small	100 km	150	2 km maritime, 5 km aviation	Above commercial airspace
Medium	50 km	300	5 km maritime, 10 km aviation	NOTAM required
Large	20 km	200	10 km maritime, 20 km aviation	NOTAM required

2.2 Fail-Safe Hazards Analysis

Primary hazard: spear impact if trajectory fails. Worst case: spear falls straight back to launch area. Time of fall from apogee (100 km): $t = \sqrt{2h/g} = \sqrt{20400/9.81} = 144 \text{ s}$ Warning time: 144 seconds from abort decision to impact — adequate for vessel evacuation.

Kinetic energy on impact (worst case, Scale 3, 10 kg spear falls 100 km): $KE = mgh = 10 * 9.81 * 100000 = 9.81 \text{ MJ}$ (equivalent to 2.35 kg TNT)

All personnel must maintain 2 km exclusion zone from projected fall zone during flight.

3. Dive Safety

Divers attaching ropes to submerged spear: - Maximum rope attachment depth: 5 m (diver safety limit for breath-hold; SCUBA for deeper) - But rope attachment at 30m depth requires SCUBA with redundant equipment - Alternative: ROV (Remotely Operated Vehicle) handles rope attachment at all depths - ROV: SeaDronics HD-1 (BFRP/aluminum hybrid, small enough for RIB deployment): ~\$8,000-\$15,000

Citations (Vol 24)

- United Nations, UNCLOS — United Nations Convention on the Law of the Sea, 1982
- 14 CFR Part 450, FAA Launch and Reentry Regulations, 2021
- ITU Radio Regulations, Edition of 2020
- USCG, COMDTPUB P16502.5, “Procedures for Filing NOTMAR,” 2022
- IMCA D 014, “Guidance for Diving Supervisors,” 2016

End Vol 24 / CSMFAB000000000113 / CSM Safe Pod Engineering Company / June 2026